Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/03/2018
PI Name:	Simpson, Richard Ph.D.		
Project Title:	Effects of Long-Term Exposure to Microgravity on Salivary Markers of Innate Immunity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rjsimpson@email.arizona.edu	**Fax:**	FY
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City:	Tucson	**State:**	AZ
Zip Code:	85721-0001	**Congressional District:**	3
Comments:	NOTE: Formerly at University of Houston until September 2017 move to University of Arizona.		
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	11/03/2011	**End Date:**	12/14/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Vos, Jessica	**Contact Phone:**	
Contact Email:	jessica.r.vos@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Flight Definition phase NOTE: End date changed to 12/14/2018 per NSSC information (Ed., 10/1/18) NOTE: End date changed to 5/2/2018 per NSSC information (Ed., 11/22/17) NOTE: End date changed to 11/2/2017 per NSSC information (Ed., 1/23/17) NOTE: End date changed to 11/2/2016 per NSSC information (Ed., 7/17/15) NOTE: Gap Immune05 deleted per IRP Rev E (Ed., 3/24/14)		
Key Personnel Changes/Previous PI:	September 2018 report: Dr. Satish Mehta has joined the project as CoInvestigator.		
COI Name (Institution):	Clarke, Mark Ph.D. (University of Houston) Crucian, Brian Ph.D. (Wyle Laboratories, Inc.) O'Connor, Dan Ph.D. (University of Houston) Pierson, Duane Ph.D. (NASA Johnson Space Center) Spielmann, Guillaume Ph.D. (University of Houston) Mehta, Satish Ph.D. (NASA Johnson Space Center)		

Grant/Contract No.:	NNX12AB48G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Immune system dysregulation has been documented during and after spaceflight, but it is not known if these changes increase infection susceptibility or pose a significant health risk to crewmembers. Inherent problems with current in-flight research are small sample sizes and the difficulty to control for the many confounding factors that impact on the immune system. As such, it is not known if changes in immunity are due to the microgravity environment per se, or to the stressors associated with landing and re-adaptation to the 1G environment. The present project proposes a Flight Definition investigation, utilizing a longitudinal repeated measures design to determine the effects of long-term exposure to microgravity on a host of salivary antimicrobial proteins (AMPs) associated with innate host immune defense, whilst also considering the impact of other acute stressors such as launch, Soyuz landing, and extravehicular activity (EVA). Saliva samples will be collected from crewmembers selected for International Space Station (ISS) mission and ground-based controls at bi-weekly intervals for 6 months prior to flight, during the 6-month period on the ISS, and for 1 month on return to Earth. Saliva sampling was selected because it is an excellent biological fluid with which to detect broad-spectrum biomarkers of front-line host immune defense and is suitable for the spaceflight environment. Attempts will also be made to establish relationships between AMPs and other stressors associated with spaceflight (i.e., mood state disturbances, circadian desynchronization, sleep loss/disruption, stress biomarkers) using serial data. Finally, blood samples will be collected before and after the mission to determine the impact of spaceflight on cellular aspects of innate immunity. Given the potential of salivary AMPs to serve as an indicator of weakened immunity during spaceflight, this project will serve as a foundation for future countermeasure developments and technological advances to detect real time changes during subsequent lunar or Mars missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will improve our understanding on how acute and long-term stress impacts on multiple aspects of the immune system. These research findings will be useful to determine if any immune related health problems might exist in individuals exposed to stressful environments (i.e., soldiers, caregivers).
Task Progress:	Project objectives for the last reporting period were: 1. Complete all data collection procedures for 8 crewmembers (including 1y mission) and 7 ground-based control subjects. 2. Continue analysis of archived samples for all enrolled subjects. 3. Disseminate research findings to the scientific community. Overview of Work Completed Study Progress: The study was initiated in September 2012 and data collection started in March 2013. We successfully enrolled the required number of subjects giving us a sample size of eight crewmembers and seven ground-based controls including an additional crewmember who completed a 1-year mission to the ISS. Baseline blood, urine, and saliva samples have been collected from all crewmembers and ground-based control subjects. All crewmembers and ground-based controls have completed all experimental procedures. We have obtained in flight samples and ambient blood returns from all enrolled crewmembers. All baseline and in flight ambient blood samples were processed and analyzed successfully. Frozen saliva, urine, and plasma samples have all been processed and analyzed. The IRB (Institutional Review Board) protocol was renewed in July 2018. During the last reporting period, the project team have been analyzing and interpreting data in preparation for dissemination to the scientific community. Data analysis and write up was somewhat delayed due to the Principal Investigator (PI) Dr. Simpson changing institutions (moving from the University of Houston to the University of Arizona) within the last reporting period. One scientific paper was published during the last reporting period, two others have been submitted for publication, and two others are currently in preparation (details below). One of the recent articles from this work “Long-duration spaceflight impairs NK-cell function” is currently under review with the Journal of Applied Physiology. Publications and Presentations: Our validation work for this project allowed us to assess NK-cell function in the context of latent cytomegalovirus infection, age, and exercise. This led to publications by Bigley et al. (2016) in the journals Cellular Immunology and Clinical and Experimental Immunology, Bigley et al. (2015) in Oxidative Medicine and Cellular Longevity and Brain, Behavior, and Immunity, and Bigley et al. (2018) in Clinical and Experimental Immunology. We also published a manuscript in European Journal of Applied Physiology, which stemmed from the validation work of our saliva assays for this project (Kunz et al., 2015). In addition, Spielmann et al. (2016) and Kunz et al. (2018) used the validation work from the viral T-cell quantification component of the “Salivary Markers” study for a publication in Scientific Reports and Physiology and Behavior, respectively. Furthermore, Graff et al. (2018), Agha et al. (2018), and LaVoy et al. (2017) used the validation work for cellular phenotyping of the “Salivary Markers” study in their recent publications in Brain, Behavior, and Immunity and Physiological Reports. The PI (Dr. Simpson) collaborated with NASA scientists and other space life scientists to generate two review articles pertaining to the effects of space travel on immune function (Crucian et al. 2018; Makedonas et al., 2018). Hematology data from this project was also used in the BMC Hematology publication by Kunz et al. (2017). NASA funding was acknowledged in all of these publications. The work supported by this research grant was presented at the International Society of Exercise and Immunology (ISEI) 2017 Symposium in July 2017, the Human Research Program Investigators' Workshop in January 2017 and February 2018, and the American College of Sports Medicine (ACSM) annual meetings in 2017 and 2018. Current and Future Work All data has been collected and analyzed. The following papers are a direct result of this project and have been published (See also Cumulative Bibliography): Crucian BE, Chouker A, Simpson RJ, Mehta S, Marshall G, Smith SM, Zwart SR, Heer M, Ponomarev S, Whitmire A, Frippiat JP, Douglas GL, Lorenzi H, Buchheim JI, Makedonas G, Ginsburg GS, Ott CM, Pierson DL, Krieger SS, Baecker N & Sams C (2018). Immune System Dysregulation During Spaceflight: Potential Countermeasures for Deep Space Exploration Missions. Frontiers in Immunology, 9, 1437. Crucian, B., Simpson, R.J., Mehta, S., Stowe, R., Chouker, A., Hwang, S., Actor, J.K., Salam, A.P., Pierson, D. and C. Sams. (2014) Terrestrial stress analogs for spaceflight associated immune system dysregulation. Brain, Behavior and Immunity, 39, 23-32. Kunz H, Quiriarte H, Simpson RJ, Ploutz-Snyder R, McMonigal K, Sams C & Crucian B. (2017). Alterations in

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Bibliography Type:	Description: (Last Updated: 01/12/2026)
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